

ICE-5121: Advance Machine Learning

Dr. Mohammad Amzad Hossain

Associate Professor

Department of Information & Communication Engineering

Noakhali Science & Technology University

Decision Tree

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- ❑ Decision Tree is a Supervised learning technique that can be used for classification problems.
- ❑ It is called a decision tree because, similar to a tree, it starts with the root node, which expands on further branches and structure like as a tree.
- ❑ In a Decision tree, there are two nodes, which are the **Decision Node** and **Leaf Node**.

Decision Tree

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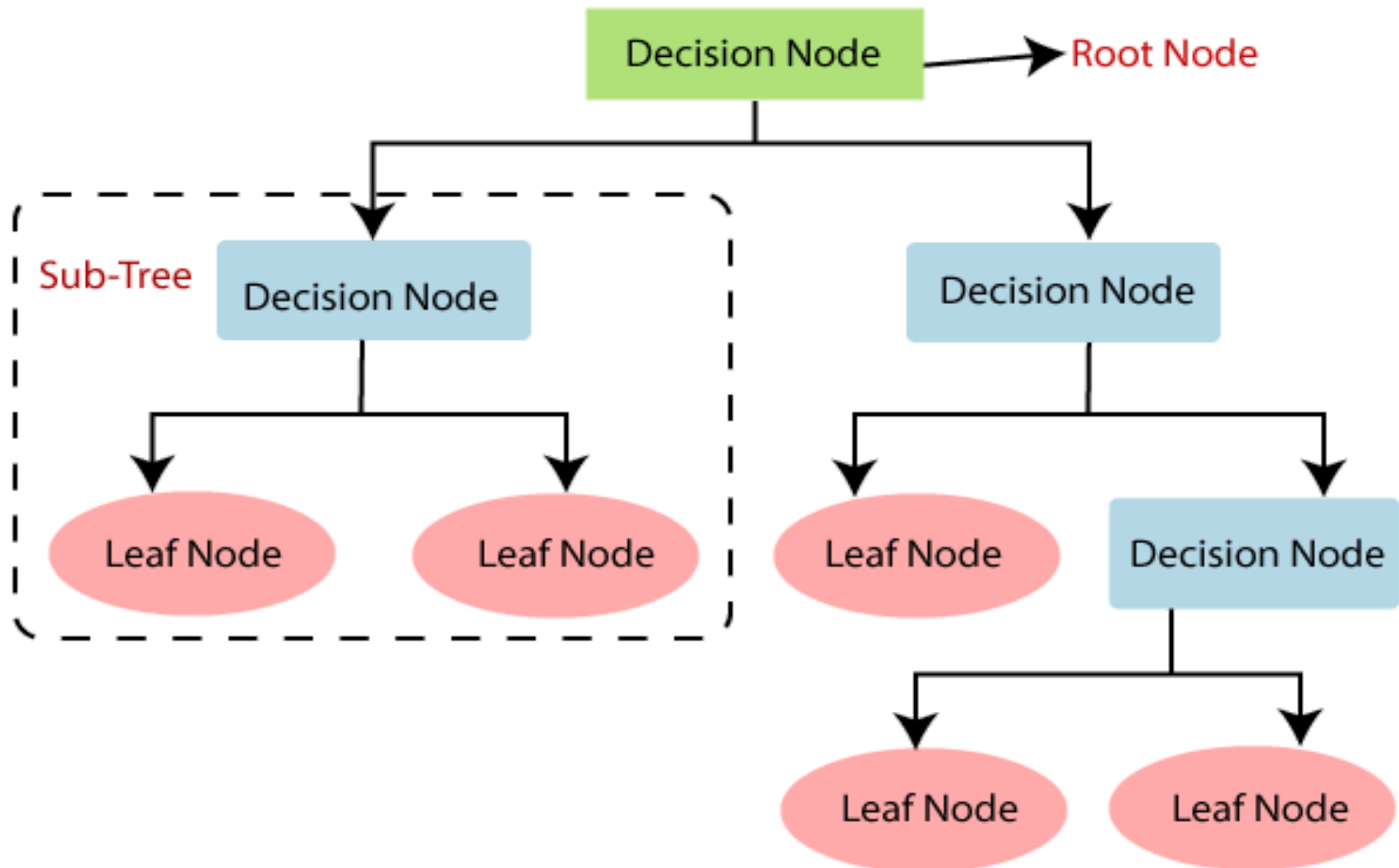


Fig. 1: Decision Tree

Why use Decision Trees?

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Below are the three reasons for using the Decision tree:

- ❑ Decision Trees usually mimic human thinking ability while making a decision, so it is easy to understand.
- ❑ The logic behind the decision tree can be easily understood because it shows a tree-like structure.
- ❑ It can handle both numerical and categorical data. It can also handle multi-output problems.

Decision Tree Algorithms

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- ❑ **Iterative Dichotomiser 3 (ID3):** This algorithm uses Information Gain to decide which attribute is to be used to classify the current subset of the data. For each level of the tree, information gain is calculated for the remaining data recursively.
- ❑ **C4.5:** This algorithm is the successor of the ID3 algorithm. This algorithm uses either Information gain or Gain ratio to decide upon the classifying attribute. It is a direct improvement from the ID3 algorithm as it can handle both continuous and missing attribute values.
- ❑ **Classification and Regression Tree (CART):** It is a dynamic learning algorithm which can produce a regression tree as well as a classification tree depending upon the dependent variable.

Steps in ID3 Algorithm

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- ❑ It begins with the original dataset S .
- ❑ On each iteration of the algorithm, it iterates through the every unused attribute of the dataset S and calculates Entropy (H) and Information gain (IG) of this attribute.
- ❑ It then selects the attribute which has the smallest Entropy or Largest Information gain.
- ❑ The dataset S is then split by the selected attribute (Root Node) to produce a subset of the data.
- ❑ The algorithm continues to recur on each subset, considering only attributes never selected before.
- ❑ Repeat until we run out of all attributes, or the decision tree has all leaf nodes.

Attribute Selection Measures

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- ❑ If the dataset consists of N attributes then deciding which attribute to place at the root or at different levels of the tree as internal nodes is a complicated step. By just randomly selecting any node to be the root can't solve the issue.
- ❑ For solving this attribute selection problem, researchers worked and devised some solutions. They suggested using some criteria like : (i) **Entropy** (ii) **Information gain** (iii) **Gini Index**

Entropy

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Entropy is a measure of the randomness in the information being processed. The higher the entropy, the harder it is to draw any conclusions from that information. Flipping a coin is an example of an action that provides information that is random.

Consider a dataset with N classes. The entropy may be

$$E = - \sum_{i=1}^N p_i \log_2 p_i$$

p_i is the probability of randomly selecting an example in class i .

Information Gain

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We can define information gain as a measure of how much information a feature provides about a class. Information gain helps to determine the order of attributes in the nodes of a decision tree.

$$\text{Information Gain} = \text{entropy}(\text{parent}) - [\text{average entropy}(\text{children})]$$

Dataset

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Outlook	Temp	Humidity	Windy	Play Golf
Rainy	Hot	High	False	No
Rainy	Hot	High	True	No
Overcast	Hot	High	False	Yes
Sunny	Mild	High	False	Yes
Sunny	Cool	Normal	False	Yes
Sunny	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Rainy	Mild	High	False	No
Rainy	Cool	Normal	False	Yes
Sunny	Mild	Normal	False	Yes
Rainy	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes

Entropy Calculation for One Attribute

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- Mathematically Entropy for one attribute is represented as:

$$E(S) = \sum_{i=1}^c -p_i \log_2 p_i$$

Play Golf	
Yes	No
9	5



Entropy(PlayGolf) = Entropy (5,9)
= Entropy (0.36, 0.64)
= - (0.36 log₂ 0.36) - (0.64 log₂ 0.64)
= 0.94

Entropy Calculation for Multiple Attribute

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- Mathematically Entropy for multiple attributes is represented as:

$$E(T, X) = \sum_{c \in X} P(c) E(c)$$

		Play Golf		
		Yes	No	
Outlook	Sunny	3	2	5
	Overcast	4	0	4
	Rainy	2	3	5
				14

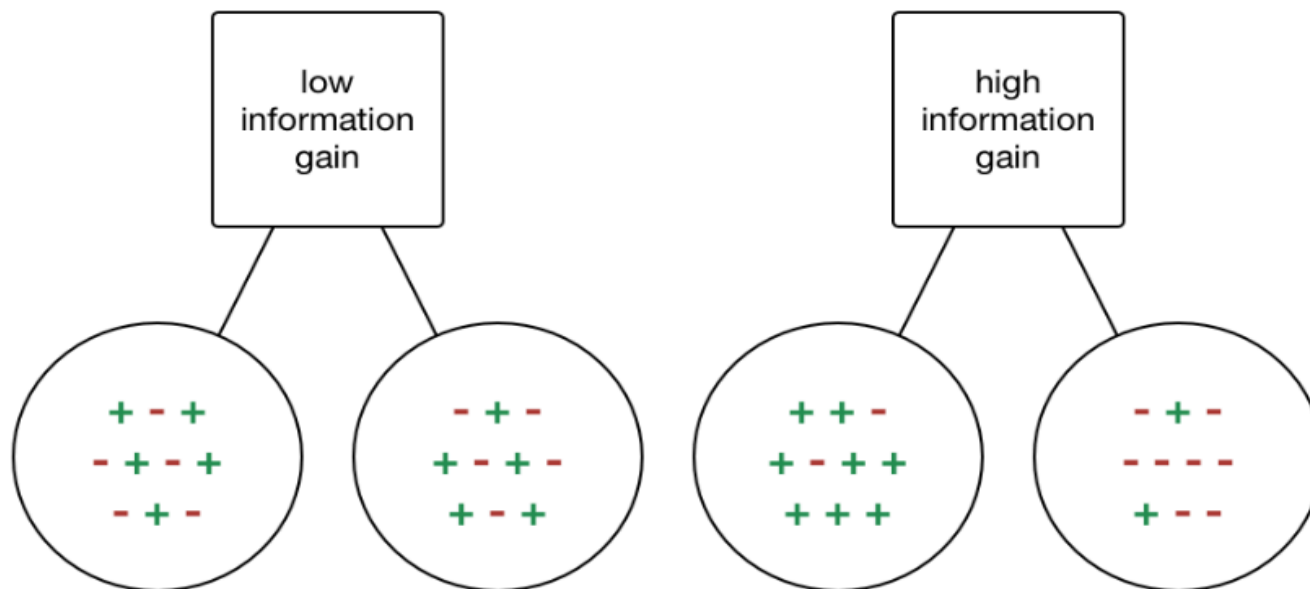


$$\begin{aligned} E(\text{PlayGolf}, \text{Outlook}) &= P(\text{Sunny}) * E(3,2) + P(\text{Overcast}) * E(4,0) + P(\text{Rainy}) * E(2,3) \\ &= (5/14) * 0.971 + (4/14) * 0.0 + (5/14) * 0.971 \\ &= 0.693 \end{aligned}$$

Information Gain (IG)

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- We can define information gain as a measure of how much information a feature provides about a class. Information gain helps to determine the order of attributes in the nodes of a decision tree. Constructing a decision tree is all about finding an attribute that returns the highest information gain and the smallest entropy.



Information Gain Calculation

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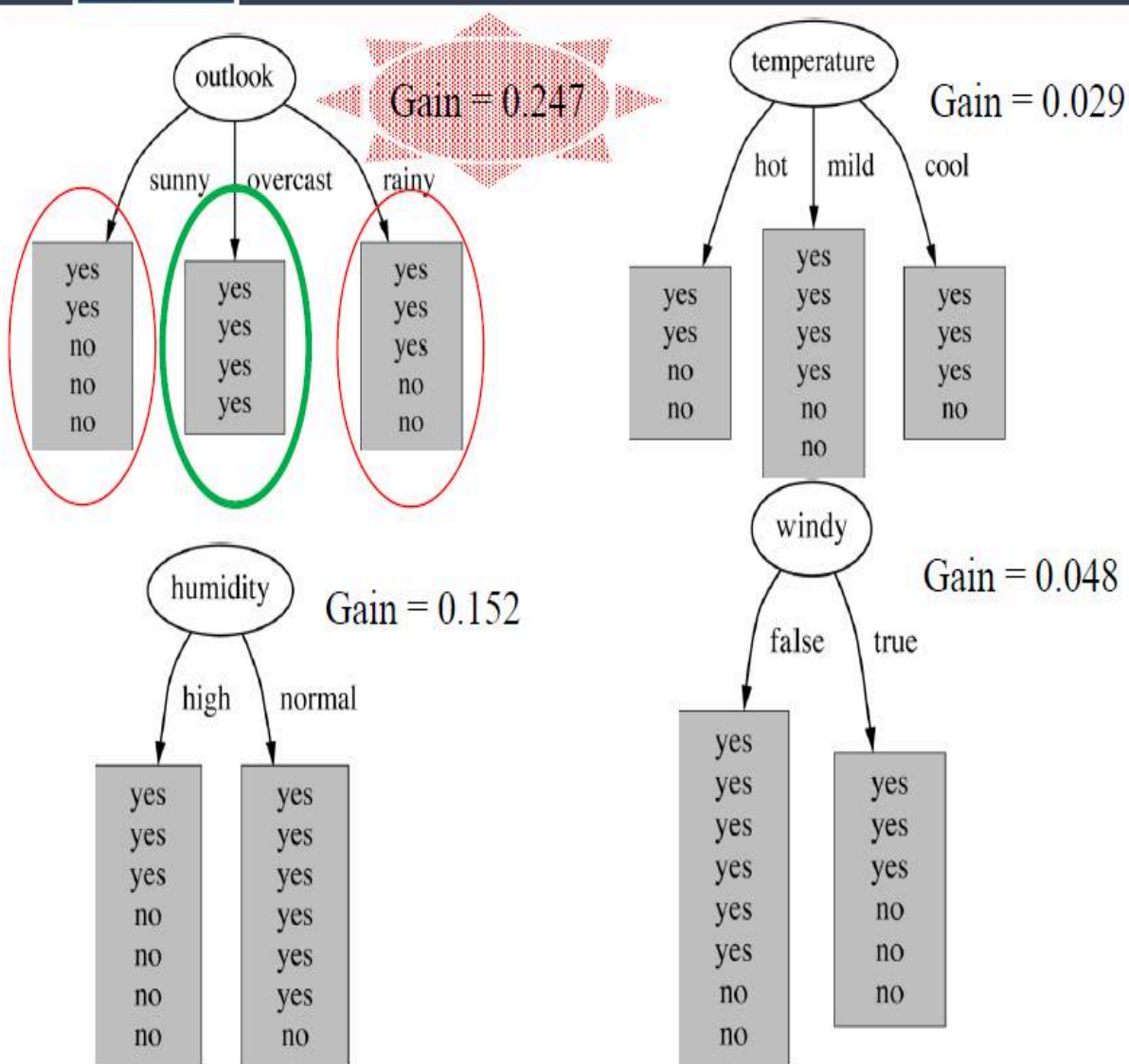
- Mathematically, IG is represented as:

$$\text{Information Gain}(T, X) = \text{Entropy}(T) - \text{Entropy}(T, X)$$

$$\begin{aligned}\text{IG}(\text{PlayGolf}, \text{Outlook}) &= E(\text{PlayGolf}) - E(\text{PlayGolf}, \text{Outlook}) \\ &= 0.940 - 0.693 \\ &= 0.247\end{aligned}$$

Drawing Decision Tree

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Anyone for Tennis?

ID	Outlook	Temp	Humidity	Windy	Play?
A	sunny	hot	high	false	no
B	sunny	hot	high	true	no
C	overcast	hot	high	false	yes
D	rainy	mild	high	false	yes
E	rainy	cool	normal	false	yes
F	rainy	cool	normal	true	no
G	overcast	cool	normal	true	yes
H	sunny	mild	high	false	no
I	sunny	cool	normal	false	yes
J	rainy	mild	normal	false	yes
K	sunny	mild	normal	true	yes
L	overcast	mild	high	true	yes
M	overcast	hot	normal	false	yes
N	rainy	mild	high	true	no

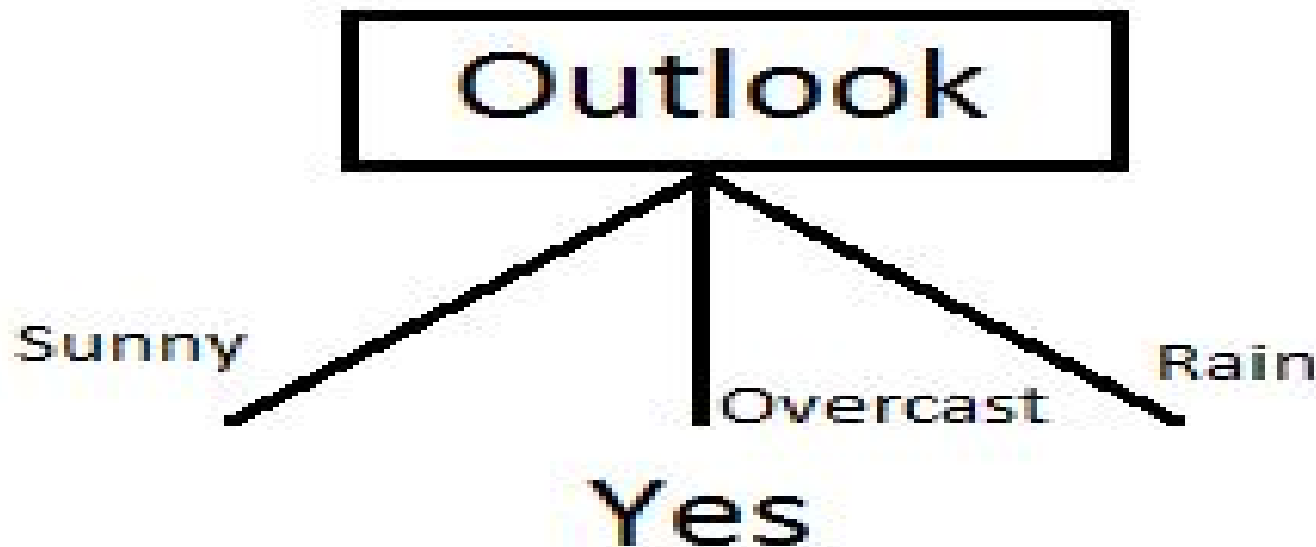
Having found the best split for the root node, repeat the whole procedure with each subset of examples ...

S will now refer to the subset in the partition being considered, instead of the entire dataset

Drawing Decision Tree (cont.)

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- Now select the feature having the largest Information gain. Here it is Outlook. So it forms the first node(root node) of our decision tree.



Drawing Decision Tree (cont.)

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Now our data look as follows :

Outlook	Temperature	Humidity	Wind	Played football(yes/no)
Sunny	Hot	High	Weak	No
Sunny	Hot	High	Strong	No
Sunny	Mild	High	Weak	No
Sunny	Cool	Normal	Weak	Yes
Sunny	Mild	Normal	Strong	Yes

Outlook	Temperature	Humidity	Wind	Played football(yes/no)
Overcast	Hot	High	Weak	Yes
Overcast	Cool	Normal	Strong	Yes
Overcast	Mild	High	Strong	Yes
Overcast	Hot	Normal	Weak	Yes

Outlook	Temperature	Humidity	Wind	Played football(yes/no)
Rain	Mild	High	Weak	Yes
Rain	Cool	Normal	Weak	Yes
Rain	Cool	Normal	Strong	No
Rain	Mild	Normal	Weak	Yes
Rain	Mild	High	Strong	No

Drawing Decision Tree (cont.)

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- Calculate parent entropy $E(\text{sunny})$

Outlook	Temperature	Humidity	Wind	Played football(yes/no)
Sunny	Hot	High	Weak	No
Sunny	Hot	High	Strong	No
Sunny	Mild	High	Weak	No
Sunny	Cool	Normal	Weak	Yes
Sunny	Mild	Normal	Strong	Yes

- $E(\text{sunny}) = (-(3/5)\log(3/5) - (2/5)\log(2/5)) = 0.971$.
- Now Calculate the information gain of Temperature.
 $IG(\text{sunny}, \text{Temperature})$

		play		total
		yes	no	
Temperature	hot	0	2	2
	cool	1	1	2
	mild	1	0	1
				5

Drawing Decision Tree (cont.)

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- ❑ $E(\text{sunny, Temperature}) = (2/5)*E(0,2) + (2/5)*E(1,1) + (1/5)*E(1,0) = 2/5 = 0.4$

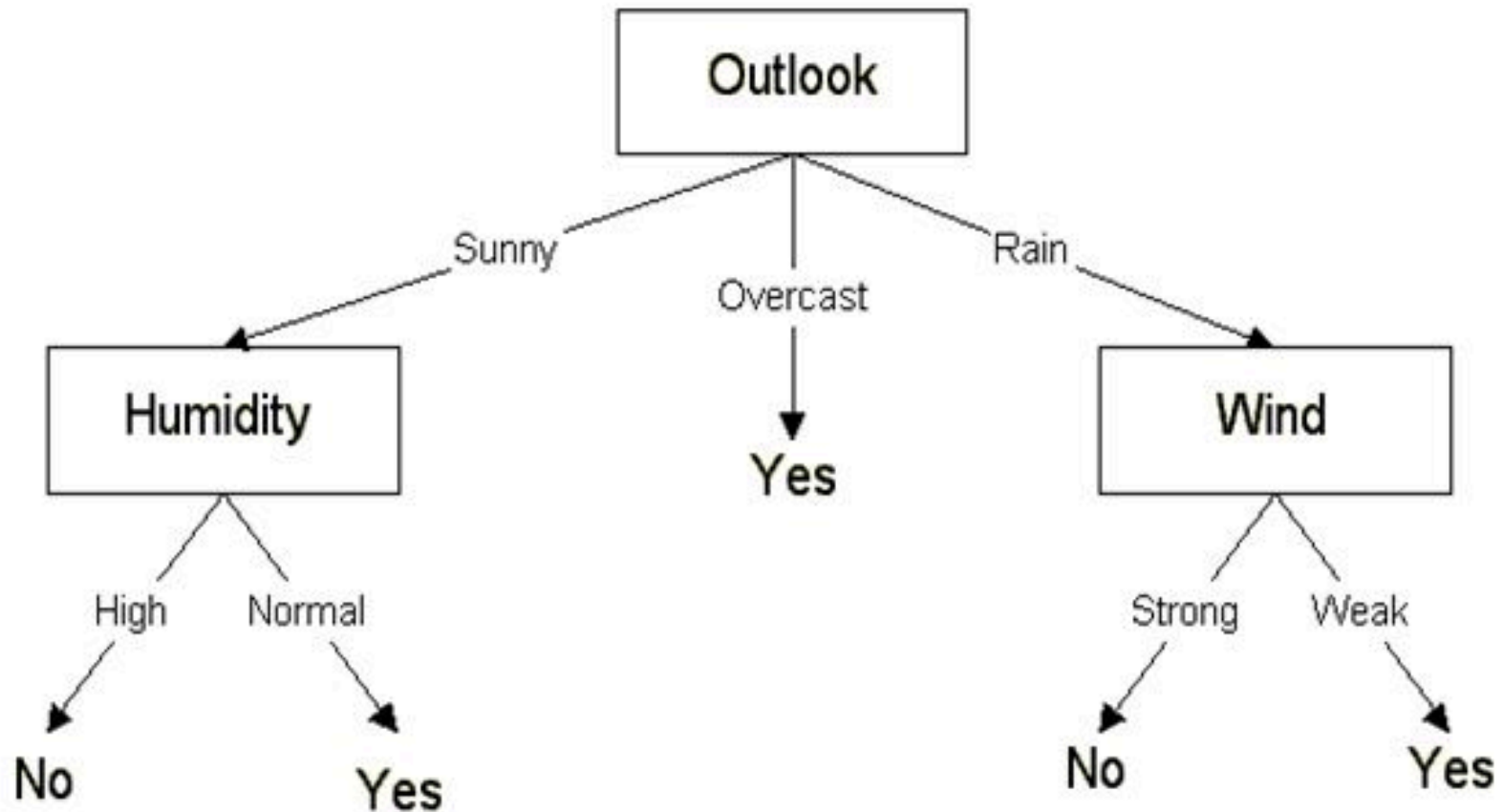
		play		total	
		yes	no		
Temperature	hot		0	2	2
	cool		1	1	2
	mild		1	0	1
					5

- ❑ Now calculate information gain.
- ❑ $IG(\text{sunny, Temperature}) = 0.971 - 0.4 = 0.571$
- **Similarly we get**
- ❑ $IG(\text{sunny, Humidity}) = 0.971$
- ❑ $IG(\text{sunny, Windy}) = 0.020$
- ❑ Here $IG(\text{sunny, Humidity})$ is the largest value. So Humidity is the node that comes under sunny. Similarly, find the nodes under rainy.

Drawing Decision Tree (cont.)

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- Finally, our decision tree will look as below:



ID3 Algorithm

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❑ Finally, our decision tree will look as below:

```
1. ID3(Examples, Attributes, Target):  
2. Input: Examples: set of classified examples  
3.       Attributes: set of attributes in the examples  
4.       Target:    classification to be predicted  
5. if Examples is empty then return a Default class  
6. else if all Examples have same class then return this class  
7. else if all Attributes are tested then return majority class  
8. else:  
9.   let Best = attribute that best separates Examples relative to Target  
10.  let Tree = new decision tree with Best as root node  
11.  foreach value  $v_i$  of Best:  
12.    let Examplesi = subset of Examples that have Best= $v_i$   
13.    let Subtree = ID3(Examplesi, Attributes-Best, Target)  
14.    add branch from Tree to Subtree with label  $v_i$   
15.  return Tree
```

BASE CASES

RECURSIVE CALL

Home work

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- ❑ **Classification using CART algorithm:** Classification using CART is similar to ID3. But instead of entropy, we use **Gini index**.

$$Gini = 1 - \sum_{i=1}^c (p_i)^2$$